Notes: Goldin and Rouse (AER, 2000)

Orchestrating Impartiality: The Impact of "Blind" Auditions on Female Musicians

Contents

1 Big picture

1.1 Research question

Goldin and Rouse study whether concealing an applicant's identity during orchestra auditions changed the hiring outcomes of female musicians. The empirical setting is unusually clean because the work product being evaluated is a short musical performance, and the treatment is a physical screen that prevents the audition committee from seeing the candidate.

The central question is: when evaluators cannot observe the candidate's sex, are women more likely to advance through auditions and to be hired into major U.S. symphony orchestras?

1.2 Main answer

The weight of the evidence is yes. The screen increases the probability that a woman advances in some audition rounds and increases the probability that a woman is hired, although several estimates are imprecise because the number of fully blind final rounds is small. The strongest evidence comes from specifications with individual fixed effects, which compare the same musician's outcomes in blind and not-blind rounds.

The results imply that blind auditions helped reduce sex-biased evaluation and contributed meaningfully to the rise in the share of women among new orchestra hires.

1.3 Why the setting is powerful

The paper exploits a change in hiring procedure rather than relying only on wage gaps or representation differences. Orchestras adopted blind auditions at different times and in different rounds. Some rounds were blind, some were not, and some musicians auditioned multiple times under both regimes. This generates variation across orchestras, time, rounds, and individuals.

The most valuable feature of the data is that the authors observe the full audition process for many auditions: who entered, who advanced, who reached later rounds, and who won. They can link musicians across auditions and use individual fixed effects to absorb time-invariant ability.

1.4 Main empirical message

Raw comparisons can be misleading. In the simple tabulations, women sometimes look less successful in blind auditions because the screen appears to have expanded the pool of women willing to audition, including more marginal candidates. Once the authors compare the same musician across blind and not-blind rounds, the screen generally improves women's outcomes.

The largest point estimates appear in preliminary rounds without semifinals and in final rounds. The semifinal round is the main anomaly: the estimated female-by-blind interaction is negative there.

2 Incremental contribution of the paper

2.1 Relative to discrimination research

Most empirical discrimination work studies earnings gaps or audit-study callbacks. Goldin and Rouse study actual hiring decisions in high-stakes jobs, using the internal records of elite orchestras. The design brings the analysis closer to the moment at which discrimination may occur: the evaluation and selection of candidates.

2.2 Relative to occupational segregation work

A large literature asks whether male-female occupational differences reflect preferences, productivity, training, or discrimination. This paper provides direct evidence that institutional procedures can affect representation in an elite occupation. The rise of women in orchestras is not explained solely by more women graduating from music schools.

2.3 Relative to policy and organizational design

The paper shows how a low-cost procedural change can affect high-skill hiring. Blind auditions are analogous to anonymized review procedures: they do not change the underlying supply of candidates or the objective content of the performance, but they change what evaluators can observe.

2.4 Relative to causal inference

The paper is an early and influential example of using institutional variation and fixed effects to study discrimination. The key comparison is not simply women versus men, or blind versus not blind, but the interaction of female status with blind audition conditions, controlling for individual musician fixed effects.

3 Institutional background

3.1 Orchestra labor market

Major U.S. orchestras have roughly fixed size and standardized positions. This is useful empirically because changes in the share of women cannot be explained by changes in the number of jobs or by the expansion of female-dominated sections. An orchestra cannot dramatically expand its headcount or alter its section structure without changing its musical function.

The jobs are also valuable and stable. Once hired, musicians often receive tenure after a probationary period. This means that the flow of new hires is small relative to the stock of orchestra members, so even modest changes in the hiring probability of women can have large long-run effects on the composition of orchestras.

3.2 Old and new audition procedures

Before the reforms, candidates were often known to the music director or to section principals. The process became more open in the 1970s and 1980s: positions were advertised, more candidates entered, committees replaced unilateral conductor discretion, and auditions became more routinized.

The screen was introduced to hide the candidate's identity from the committee. The screen was most commonly used in preliminary rounds, sometimes used in semifinals, and rarely used in finals. Some orchestras also used carpets or other procedures to mask footsteps.

3.3 Why sex could matter

The paper documents the historical context: women were rare in elite orchestras, and some prominent conductors expressed views that women were inferior orchestral musicians. In such a setting, a visible audition may allow evaluators' beliefs about sex to enter the selection decision. A blind audition removes that channel.

4 Data

4.1 Roster data

The roster data come from the personnel pages of orchestra programs for 11 major U.S. symphony orchestras. These data identify musicians, instruments, positions, and the year in which musicians appear in the orchestra. They are used to track the share of women in orchestras and among new hires.

The roster sample covers the long-run evolution of orchestra composition and is especially useful for estimating how blind audition regimes are associated with the sex of newly hired musicians from 1970 to 1996.

4.2 Audition records

The audition records come from eight major orchestras. These records list candidates, the round in which they participated, whether they advanced, and who was ultimately hired. The authors record instrument, position, round type, whether the round was blind, and candidate sex.

The final analysis sample includes thousands of person-rounds. The paper reports 7,065 individuals, 588 audition rounds, and 14,121 person-round observations after applying sample restrictions. About 24 percent of individuals appear in more than one audition, which is crucial for the individual fixed-effects design.

4.3 Why individual fixed effects matter

If blind auditions attract a broader and potentially less preselected pool of women, raw blind-versus-not-blind comparisons may understate the benefit of the screen. Individual fixed effects control for time-invariant musician quality. The central coefficient then compares the same musician's outcomes under blind and not-blind conditions.

5 Econometric framework

5.1 Baseline selection equation

Let P_{ijtr} denote whether musician i advances in orchestra j, year t, round r. The paper's conceptual model is

$$P_{ijtr} = f(X_{it}, F_i, B_{jtr}, Z_{jtr}),$$

where F_i indicates that the musician is female, B_{jtr} indicates that the round is blind, X_{it} are individual controls, and Z_{jtr} are orchestra-round controls.

The linear version is

$$P_{ijtr} = \alpha + \beta F_i + \gamma B_{jtr} + \delta(F_i \times B_{jtr}) + X_{it}\theta + Z_{jtr}\phi + \varepsilon_{ijtr}.$$

The parameter of interest is δ. It measures whether the screen changes women's advancement probability relative to men's advancement probability.

5.2 Individual fixed-effects version

The most informative specification adds musician fixed effects:

$$P_{ijtr} = \alpha_i + \gamma B_{jtr} + \delta(F_i \times B_{jtr}) + X_{it}\theta + Z_{jtr}\phi + \varepsilon_{ijtr}.$$

Because F_i is time-invariant, it is absorbed by α_i. Identification comes from musicians observed in both blind and not-blind conditions. In this specification, δ compares how the screen changes outcomes for women relative to men, holding constant each individual's fixed ability component.

5.3 Hiring equation

For hiring, the paper constructs whether an audition is completely blind. The equivalent equation uses an indicator for a fully blind audition and its interaction with female status:

$$H_{ij} = \alpha_i + \gamma C_j + \delta(F_i \times C_j) + W_{ij}\psi + u_{ij},$$

where $H_{ij} = 1$ if musician i is hired in audition j, and $C_j = 1$ if the audition is completely blind. Again, δ is the incremental effect for women.

5.4 Roster probit equation

The roster analysis estimates whether a new hire is female as a function of the orchestra's blind-audition regime. A simplified version is

$$\Pr(\text{Female new hire}_{jtp} = 1) = \Phi(\alpha_j + \lambda_j t + \eta \, \text{BlindRegime}_{jt} + \text{section fixed effects}).$$

The roster regressions ask whether women are more likely to enter orchestras after the orchestra adopts blind procedures, controlling for orchestra fixed effects and orchestra-specific time trends.

6 Core findings

6.1 Trends in female representation

The figures show a dramatic increase in female representation beginning around the 1970s and 1980s. Among the Big Five orchestras, no orchestra had more than about 12 percent women until roughly 1980. By the 1990s, the share is much higher, with the New York Philharmonic reaching roughly 35 percent.

The share of women among new hires rises even more sharply. Before 1970, women were less than 10 percent of new hires in the Big Five. In later decades, women often account for about one-third to one-half of new hires.

The Juilliard figure weakens a pure supply-side explanation. The share of female graduates does rise over the long run, but there is no sharp break that lines up perfectly with the sudden increase in female hires.

6.2 Raw audition success

Raw tabulations are ambiguous. Table 4 suggests that women appear less successful in blind auditions in the full sample. But this comparison mixes changes in the candidate pool with changes in evaluation. If the screen encouraged more women to enter, the average quality of female entrants in blind auditions may have changed.

Table 5 therefore restricts attention to musicians who auditioned both blind and not blind. In that subset, women do better in blind preliminary rounds without semifinals and better in blind finals. The improvement is economically large relative to the baseline success rates.

6.3 Fixed-effects advancement estimates

Table 6 is the main individual fixed-effects evidence by round. In preliminary rounds without semifinals, the female-by-blind estimate is about 0.11 to 0.13, implying an 11 to 13 percentage point relative improvement for women. In final rounds, the estimate is about 0.31 to 0.33. These are large point estimates, though final-round estimates have large standard errors because blind finals are rare.

The semifinal estimates are negative. The authors treat this as a persistent anomaly. A plausible interpretation is that preliminary rounds and semifinals served different filtering roles, and semifinals may have involved different committee behavior or sample composition.

6.4 Addressing confounds

Table 7 shows that adding orchestra fixed effects does not remove the positive female-by-blind effect in the all-round specification. Table 8 addresses sex misclassification using census-based probabilities of candidate sex and produces broadly similar results. Table A4 checks whether short-panel bias drives the fixed-effects results, and the conclusions are similar in magnitude.

6.5 Hiring and long-run composition

Table 9 shows that completely blind auditions increase a woman's probability of being hired mainly when there is no semifinal round. The point estimate is about 5 percentage points without year controls and about 4 percentage points with them. The effect is much smaller when semifinals are present.

Table 10 uses roster data and finds that blind audition regimes increased the probability that a new hire was female. The point estimates imply a roughly 7.5 percentage point increase in the probability that a new hire is female under any blind audition regime. Relative to the low baseline share of women among hires before 1970, this is an economically meaningful effect.

7 Interpreting the magnitude

7.1 Candidate pool versus evaluation effect

The paper separates two forces. First, the female share of the candidate pool rose over time. Second, the screen changed the probability that a woman succeeded conditional on auditioning.

Using the authors' illustrative calculation, suppose an audition has 30 candidates and one hire. In the not-blind regime, assume 20 percent of candidates are women and 10 percent of new hires are women. In the blind regime, assume 30 percent of candidates are women and 35 percent of new hires are women. The implied success rate of female candidates rises from about 1.66 percent to about 3.89 percent. The authors' estimates suggest that roughly half of this increase can be attributed to the screen.

7.2 Contribution to the rise in women among orchestras

The paper's preferred interpretation is that blind auditions can explain about one-third of the increase in the female share of new hires. The rise in the female candidate pool explains another substantial portion. Using roster data, the authors estimate that blind auditions may explain about 25 percent of the increase in the share of all orchestra musicians who are women from 1970 to 1996.

8 Identification logic and threats

8.1 Why simple before-after comparisons are not enough

Many changes occurred during the same decades as the adoption of the screen: women entered professional schools at higher rates, orchestras opened auditions more widely, committees became more democratic, and social attitudes changed. A simple increase in the female share after 1970 cannot by itself identify the effect of the screen.

8.2 How the paper improves the comparison

The paper uses variation across orchestras, rounds, and individuals. It compares blind and not-blind rounds within the same broad hiring institution and then adds individual fixed effects. This design greatly reduces the concern that women in blind auditions simply differ from women in not-blind auditions.

8.3 Remaining limitations

The final-round evidence is based on relatively few blind finals, so standard errors are large. The screen was not randomly assigned, so adoption timing may be correlated with broader institutional reforms. The authors address this by studying whether female presence predicted screen adoption and by including orchestra and time controls, but complete randomization is not available.

9 Conceptual lessons

9.1 Information control changes decisions

The paper demonstrates that changing what evaluators observe can change outcomes, even when the performance itself is unchanged. The screen removes a potentially bias-relevant signal while preserving the audition committee's access to musical performance.

9.2 Representation can change through hiring flows

Because orchestra jobs are stable and turnover is low, small changes in hiring probabilities compound slowly but meaningfully. The flow of new hires drives the eventual stock of orchestra members.

9.3 Procedural reforms may reveal latent talent

If talented women were previously discouraged or disadvantaged by visible auditions, blind auditions can both increase women's willingness to compete and improve their conditional success. The paper's raw and fixed-effects patterns are consistent with both effects.

10 Extracted figures and tables from the paper

10.1 Figure 1. Proportion Female in Nine Orchestras, 1940 to 1990s; Figure 2. Number of New Hires in Five Orchestras, 1950 to 1990s

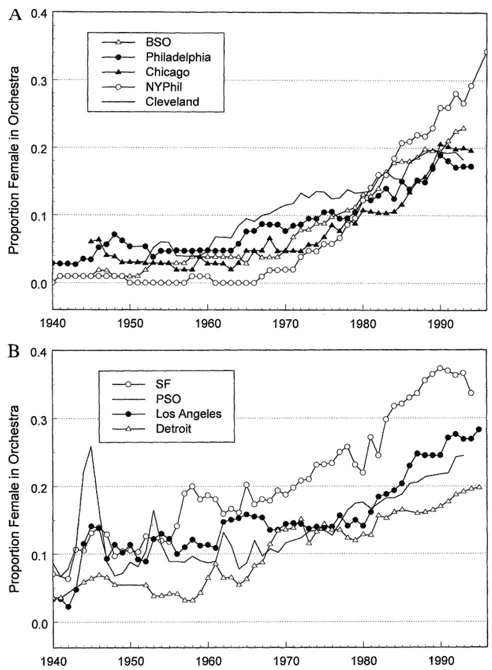


FIGURE 1. PROPORTION FEMALE IN NINE ORCHESTRAS, 1940 TO 1990's
A: THE "BIG FIVE"; B: FOUR OTHERS

Source: Roster sample. See text.

orchestras, there is a discernible pool of female contestants coming out of
r the group as a whole in the late music schools. Because the majority of new

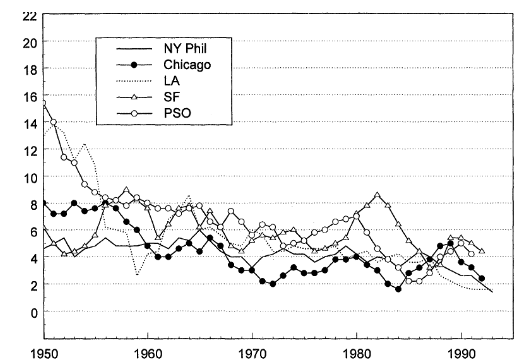


FIGURE 2. NUMBER OF NEW HIRES IN FIVE ORCHESTRAS, 1950 TO 1990's

Source: Roster sample. See text.

centered moving average is used. New hires are musicians who were not with the orchestra the previous
for at least one additional year, and who were not substitute musicians in the current year.

aks in the fraction of all gradu- sound."[11] Zubin Mehta, conductor of the Los An-
female.[10] Thus, it is not imme- geles Symphony from 1964 to 1978 and of the

Figure 1. Proportion Female in Nine Orchestras, 1940 to 1990s

Interpretation. The share of women rises slowly before the 1970s and then accelerates, especially in the New York Philharmonic and other leading orchestras. The pattern establishes the large compositional change that the paper seeks to explain.

Figure 2. Number of New Hires in Five Orchestras, 1950 to 1990s

Interpretation. New hires are few each year and trend downward over time. Because turnover is low, the large increase in the stock of female orchestra members requires a large change in the sex composition of the small flow of new hires.

10.2 Figure 3. Female Share of New Hires in Eight Orchestras, 1950 to 1990s; Figure 4. Proportion Female of Juilliard Graduates, Total and by Section, 1947 to 1995

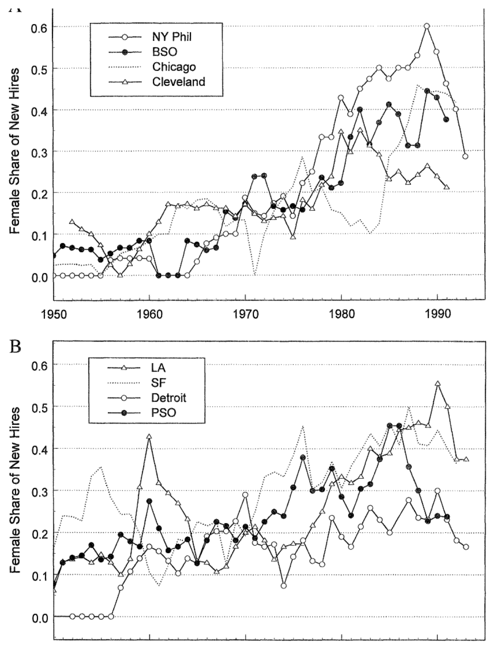
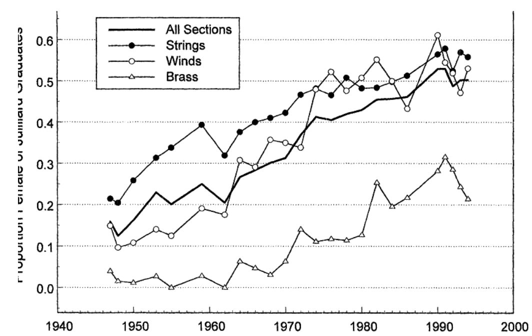


FIGURE 3. FEMALE SHARE OF NEW HIRES IN EIGHT ORCHESTRAS, 1950 TO 1990'S
A: FOUR OF THE "BIG FIVE"; B: FOUR OTHERS

Source: Roster sample. See text.

year centered moving average is used. New hires are musicians who were not with the orchestra the previous ıain for at least one additional year, and who were not substitute musicians in the current year.



URE 4. PROPORTION FEMALE OF JUILLIARD GRADUATES, TOTAL AND BY SECTION: 1947 TO 1995

Source: Juilliard Music School files.

l the world.[15] Musicians inter- itioning are required to submit a

the committee is free to advance as many, or as few, as they wish. Candidates advanced from

Figure 3. Female Share of New Hires in Eight Orchestras, 1950 to 1990s

Interpretation. Women become a much larger share of new hires after the 1970s. The movement is particularly sharp in several elite orchestras, which suggests that hiring procedures or evaluator behavior changed alongside supply-side trends.

Figure 4. Proportion Female of Juilliard Graduates, Total and by Section, 1947 to 1995

Interpretation. The supply of trained female musicians rises over the long run, but the graph does not show a sharp break sufficient to explain the timing and size of the increase in orchestra hiring by itself.

10.3 Table 1. Orchestra Audition Procedure Summary Table; Table 2. Estimated Probit Models for the Use of a Screen

rchestra	Preliminaries	Semifinals	Finals
A	Blind since 1973	Blind (varies) since 1973	Not blind
B	Blind since at least 1967	Use of screen varies	Blind 1967–1969; since winter 1994
C	Blind since at least 1979 (definitely after 1972)	Not blind: 1991–present Blind: 1984–1987	Not blind
D	Blind since 1986	Blind since 1986; varies until 1993	1st part blind since 1993; 2nd part not blind
E	Use of screen varies until 1981	Use of screen varies	Not blind
F	Blind since at least 1972	Blind since at least 1972	Blind since at least 1972
G	Blind since 1986	Use of screen varies	Not blind
H	Blind since 1970	Not blind	Not blind
I	Blind since 1979	Blind since 1979	Blind since fall 1983
J	Blind since 1952	Blind since 1952	Not blind
K	Not blind	Not blind	Not blind

otes: The 11 orchestras (A through K) are those in the roster sample described in the text. A subset of eight form the audition ımple (also described in the text). All orchestras in the sample are major big-city U.S. symphony orchestras and include the Big Five."
ources: Orchestra union contracts (from orchestra personnel managers and libraries), personal conversations with orchestra ersonnel managers, and our mail survey of current orchestra members who were hired during the probable period of screen doption.

ures employed during the audition that won columns (1) and (2) in Table 2, although the

Table 1. Orchestra Audition Procedure Summary Table

Interpretation. The table documents cross-orchestra and cross-round variation in screen adoption. This variation is the basis for the empirical design: some orchestras adopted screens earlier, some later, and some used them in preliminaries but not finals.

FOR THE USE OF A SCREEN

	Preliminaries blind		Finals blind
	(1)	(2)	(3)
(Proportion female)$_{t-1}$	2.744	3.120	0.490
	(3.265)	(3.271)	(1.163)
	[0.006]	[0.004]	[0.011]
(Proportion of orchestra personnel with <6 years tenure)$_{t-1}$	−26.46	−28.13	−9.467
	(7.314)	(8.459)	(2.787)
	[−0.058]	[−0.039]	[−0.207]
"Big Five" orchestra		0.367	
		(0.452)	
		[0.001]	
pseudo R^2	0.178	0.193	0.050
Number of observations	294	294	434

Notes: The dependent variable is 1 if the orchestra adopts a screen, 0 otherwise. Huber standard errors (with orchestra random effects) are in parentheses. All specifications include a constant. Changes in probabilities are in brackets. "Proportion female" refers to the entire orchestra. "Tenure" refers to years of employment in the current orchestra. "Big Five" includes Boston, Chicago, Cleveland, New York Philharmonic, and Philadelphia. The data begin in 1947 and an orchestra exits the exercise once it adopts the screen. The unit of observation is an orchestra-year.
Source: Eleven-orchestra roster sample. See text.

III. The Role of Blind Auditions on the

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Table 2. Estimated Probit Models for the Use of a Screen

Interpretation. The table examines whether screen adoption is predicted by orchestra composition. The results provide little evidence that a higher female share mechanically caused screen adoption, weakening the concern that adoption was simply driven by a preexisting preference for hiring women.

10.4 Table 3. Descriptive Statistics about Auditions, by Year and Round of Audition; Table 4. Average Success at Auditions by Sex, Year, and Round of Audition

Year	Number of auditions	Proportion female	Number of musicians	Number of auditions	Proportion female	Number of musicians	Number of auditions	Proportion female
			Completely blind auditions			Not completely blind auditions		
All	254	0.367 (0.013)	43.4 (3.13)	60	0.393 (0.029)	38.1 (1.74)	194	0.359 (0.015)
Pre-1970	10	0.187 (0.042)				16.3 (2.27)	10	0.187 (0.042)
1970–1979	69	0.329 (0.026)				31.4 (2.10)	69	0.329 (0.026)
1980–1989	102	0.394 (0.019)	42.5 (4.29)	33	0.375 (0.034)	39.6 (2.73)	69	0.403 (0.022)
1990+	73	0.390 (0.027)	44.6 (4.64)	27	0.415 (0.049)	50.6 (4.52)	46	0.375 (0.033)
Round				Blind rounds			Not-blind rounds	
Preliminaries, without semifinals	170	0.357 (0.015)	34.3 (1.87)	125	0.367 (0.017)	24.7 (2.33)	45	0.327 (0.029)
Preliminaries, with semifinals	137	0.396 (0.019)	45.5 (2.54)	134	0.395 (0.019)	49.3 (17.0)	3	0.425 (0.205)
Semifinals	114	0.415 (0.019)	12.3 (0.649)	89	0.404 (0.022)	10.4 (1.21)	25	0.455 (0.043)
Finals	167	0.430 (0.016)	4.93 (0.448)	28	0.472 (0.040)	7.12 (0.310)	130	0.422 (0.017)

Notes: The unit of observation for the top portion is the audition, whereas it is the round for the bottom portion (e.g., proportion female in the top portion of the table is averaged across the auditions). Standard errors are in parentheses.
Source: Eight-orchestra audition sample. See text.

the candidates played with the orchestra.[33] In addition, we generally consider each round of

candidates on average, whereas those without the screen had 26. Women were about 37

Table 3. Descriptive Statistics about Auditions, by Year and Round of Audition

Interpretation. The audition sample is concentrated after 1970. Most preliminary rounds are blind, semifinals are often blind, and finals are usually not blind. Women are a larger share of finalists than preliminary candidates, suggesting selection through the audition process.

Year	Relative female success: All auditions	Completely blind auditions	Not completely blind auditions
All	−0.001 (0.008)	−0.022 (0.012)	0.006 (0.010)
Pre-1970	0.053 (0.115)		0.053 (0.115)
1970–1979	0.001 (0.021)		0.001 (0.021)
1980–1989	−0.006 (0.009)	−0.039 (0.016)	0.010 (0.009)
1990+	−0.003 (0.010)	−0.001 (0.017)	−0.003 (0.013)
Round	All rounds	Blind rounds	Not-blind rounds
Preliminaries, without semifinals	−0.032 (0.019)	−0.048 (0.021)	0.012 (0.040)
Preliminaries, with semifinals	−0.048 (0.016)	−0.052 (0.016)	0.116 (0.228)
Semifinals	−0.030 (0.038)	−0.059 (0.044)	0.071 (0.080)
Finals	0.009 (0.036)	−0.028 (0.102)	0.016 (0.038)

Notes: For the top part of the table "success" is a "hire," whereas for the bottom portion "success" is advancement from one stage of an audition to the next. The unit of observation for the top portion is the audition, whereas it is the round for the bottom portion (e.g., relative female success in the top portion of the table is averaged across the auditions). Standard errors are in parentheses. "Relative female success" is the proportion of women advanced (or hired) minus the proportion of men advanced (or hired). By hired, we mean those who were advanced from the final round out of the entire audition.
Source: Eight-orchestra audition sample. See text.

Take the preliminary round with no semifinals, for example, in Table 5. In the blind auditions 28.6 percent of the women are advanced, as are 20.2 percent of the men. But in the not-blind column,

screen in the finals and for the overall audition (termed "hired" in the table). For the finals, a woman's success rate is increased by 14.8 percentage points moving to blind auditions

Table 4. Average Success at Auditions by Sex, Year, and Round of Audition

Interpretation. Raw relative female success is not uniformly higher in blind auditions. This is why the fixed-effects design is essential: raw differences mix the treatment effect of the screen with changes in who chooses or is allowed to audition.

10.5 Table 5. Average Success at Auditions by Sex and Stage for Musicians Who Auditioned Both Blind and Not Blind; Table 6. Linear Probability Estimates of the Likelihood of Being Advanced with Individual Fixed Effects

OF MUSICIANS WHO AUDITIONED BOTH BLIND AND NOT BLIND

	Blind: Proportion advanced	Blind: Number of person-rounds	Not blind: Proportion advanced	Not blind: Number of person-rounds
Preliminaries without semifinals				
Women	0.286 (0.043)	112	0.193 (0.041)	93
Men	0.202 (0.026)	247	0.225 (0.031)	187
Preliminaries with semifinals				
Women	0.200 (0.092)	20	0.133 (0.091)	15
Men	0.083 (0.083)	12	0.000 (0.000)	8
Semifinals				
Women	0.385 (0.061)	65	0.568 (0.075)	44
Men	0.368 (0.059)	68	0.295 (0.069)	44
Finals				
Women	0.235 (0.106)	17	0.087 (0.060)	23
Men	0.000 (0.000)	12	0.133 (0.091)	15
"Hired"				
Women	0.027 (0.008)	445	0.017 (0.005)	599
Men	0.026 (0.005)	816	0.027 (0.005)	1102

Notes: The unit of observation is a person-round. Standard errors are in parentheses. For the round in question, only musicians who auditioned more than once and who auditioned at least once behind a screen and at least once without a screen are included. "Hired" means those who were advanced from the final round out of the entire audition. Blind in the "hired" category means for all rounds. Not blind in the "hired" category means that at least one round was not blind. This difference in the definition of what constitutes a "blind" round or audition is one reason why the number of observations in the first four panels is less than the number of observations in the "hired" panel. The number of observations also differ because we exclude auditions or rounds in which no individual is advanced or in which there are only women or no women. Finally, unlike in subsequent tables, we exclude a few candidates for whom we could not determine or impute their sex. Note that the binding constraint for the preliminaries is the not-blind category, for which we have only one orchestra. The binding constraint in the "hired" category are the blind auditions, for which we have (at most) three orchestras. Musicians can appear more than once in either the blind or not-blind categories.
Source: Eight-orchestra audition sample. See text.

The results given in Table 6 are the regression analogs to the raw tabulations in Table 5.[37] Because the effect of the blind procedure could differ by the various rounds in the audition process, we divide audition rounds into the three main rounds (preliminary, semifinal, and final) and also separate the preliminaries

Table 5. Average Success at Auditions by Sex and Stage for the Subset of Musicians Who Auditioned Both Blind and Not Blind

Interpretation. Comparing musicians observed in both regimes changes the picture. Women do better in blind preliminary rounds without semifinals and in blind finals. This supports the interpretation that the screen helps women conditional on individual quality.

TABLE 6—LINEAR PROBABILITY ESTIMATES OF THE LIKELIHOOD OF BEING ADVANCED: WITH INDIVIDUAL FIXED EFFECTS

	Preliminaries: Without semifinals (1)	Preliminaries: Without semifinals (2)	Preliminaries: With semifinals (3)	Preliminaries: With semifinals (4)	Semifinals (5)	Semifinals (6)	Finals (7)	Finals (8)
Blind	−0.017 (0.039)	0.003 (0.046)	0.109 (0.172)	0.224 (0.242)	0.026 (0.089)	0.102 (0.096)	−0.154 (0.150)	−0.060 (0.149)
Female × Blind	0.125 (0.068)	0.111 (0.067)	0.013 (0.215)	−0.025 (0.251)	−0.179 (0.126)	−0.235 (0.133)	0.308 (0.196)	0.331 (0.181)
Number of auditions attended		−0.020 (0.014)		0.010 (0.010)		0.015 (0.030)		0.126 (0.028)
Years since last audition		−0.005 (0.007)		−0.006 (0.005)		−0.005 (0.013)		0.016 (0.015)
Automatic placement						−0.096 (0.064)		−0.069 (0.073)
"Big Five" orchestra		−0.154 (0.035)		−0.059 (0.024)		0.006 (0.081)		−0.059 (0.084)
Total number of auditioners in round (÷100)		−0.003 (0.081)		0.014 (0.031)		−0.371 (0.521)		−0.262 (0.756)
Proportion female at the audition round		0.118 (0.139)		0.312 (0.134)		0.104 (0.218)		0.067 (0.159)
Principal		−0.079 (0.037)		−0.078 (0.019)		−0.082 (0.066)		−0.185 (0.076)
Substitute		0.165 (0.081)		0.123 (0.093)		0.167 (0.183)		0.079 (0.217)
p-value of H_0: Blind + (Female × Blind) = 0	0.053	0.063	0.342	0.285	0.089	0.170	0.222	0.042
Year fixed effects?	No	Yes	No	Yes	No	Yes	No	Yes
R^2	0.748	0.775	0.687	0.697	0.774	0.794	0.811	0.878
Number of observations	5,395	5,395	6,239	6,239	1,360	1,360	1,127	1,127

Notes: The unit of observation is a person-round. The dependent variable is 1 if the individual is advanced to the next round and 0 if not. Standard errors are in parentheses. All specifications include individual fixed effects, an interaction for the sex being missing and a blind audition round, a dummy indicating if years since last audition is missing, and [in columns (3)–(8)] whether an automatic placement is missing.
Source: Eight-orchestra audition sample. See text.

Table 6. Linear Probability Estimates of the Likelihood of Being Advanced with Individual Fixed Effects

Interpretation. This is the main fixed-effects evidence. The female-by-blind interaction is positive and large for preliminary rounds without semifinals and for finals, but negative for semifinals. The results suggest that the screen's effect differs by round.

10.6 Table 7. Linear Probability Estimates with Individual and Orchestra Fixed Effects; Table 8. Linear Probability Estimates Addressing Sex Misclassification

TABLE 7—LINEAR PROBABILITY ESTIMATES OF THE LIKELIHOOD OF BEING ADVANCED: WITH INDIVIDUAL AND ORCHESTRA FIXED EFFECTS

	Include individual fixed effects		Exclude individual fixed effects
	(1)	(2)	(3)
Blind	0.404	0.399	0.103
	(0.027)	(0.027)	(0.018)
Female × Blind	0.044	0.041	−0.069
	(0.039)	(0.039)	(0.022)
Female			−0.005
			(0.019)
p-value of H_0: Blind + (Female × Blind) = 0	0.000	0.000	0.090
Individual fixed effects?	Yes	Yes	No
Orchestra fixed effects?	No	Yes	Yes
Year fixed effects?	Yes	Yes	Yes
Other covariates?	Yes	Yes	Yes
R^2	0.615	0.615	0.048
Number of observations	8,159	8,159	8,159

Notes: The unit of observation is a person-round. The dependent variable is 1 if the person is advanced to the next round and 0 if not. Standard errors are in parentheses. All specifications include an interaction for the sex being missing and a blind audition; "Other covariates" include automatic placement, years since last audition, number of auditions attended, size of the audition round, proportion female in audition round, whether a principal or substitute position, and a dummy indicating whether years since last audition is missing. These regressions include only the orchestras that changed their audition policy during our sample years and for which we observe individuals auditioning for the audition round both before and after the policy change. These regressions include 4,836 separate persons and are identified off of 1,776 person-rounds comprised of individuals who auditioned both before and after the policy change for a particular orchestra.
Source: Eight-orchestra audition sample (three orchestras of which are used; see Notes). See text.

Table 7. Linear Probability Estimates of the Likelihood of Being Advanced with Individual and Orchestra Fixed Effects

Interpretation. Adding orchestra fixed effects preserves a positive female-by-blind estimate in the all-round specification. This indicates that the result is not solely due to fixed differences across orchestras.

TABLE 8—LINEAR PROBABILITY ESTIMATES OF THE LIKELIHOOD OF BEING ADVANCED: ADDRESSING SEX MISCLASSIFICATION

Part A: Preliminary rounds

	Preliminaries			
	Without semifinals		With semifinals	
	OLS	IV	OLS	IV
Blind	−0.012	0.057	−0.174	0.290
	(0.043)	(0.045)	(0.093)	(0.241)
Female × Blind	0.139	0.137	0.272	−0.035
	(0.066)	(0.068)	(0.188)	(0.251)
Other covariates?	Yes	Yes	Yes	Yes
Individual fixed effects?	Yes	Yes	Yes	Yes
Year fixed effects?	Yes	Yes	Yes	Yes
R^2	0.771			
Number of observations	5,696	5,395	6,546	6,239

Part B: Semifinal and final rounds, and with orchestra fixed effects

	Semifinals		Finals		With orchestras fixed effects	
	OLS	IV	OLS	IV	OLS	IV
Blind	0.100	−0.197	−0.028	−0.025	0.010	0.061
	(0.083)	(0.700)	(0.125)	(0.141)	(0.028)	(0.033)
Female × Blind	−0.242	−0.193	0.160	0.324	0.069	0.052
	(0.120)	(0.429)	(0.171)	(0.181)	(0.035)	(0.036)
Other covariates	Yes	Yes	Yes	Yes	Yes	Yes
Individual fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
R^2	0.776		0.848		0.654	
Number of observations	1,600	1,360	1,509	1,127	8,882	8,159

Notes: The unit of observation is a person-round. The dependent variable is 1 if the individual is advanced to the next round and 0 if not. Standard errors are in parentheses. The instruments are the census probability that the individual is female, a dummy for whether the person has been sexed with certainty, and proportion female calculated using the census data and an interaction between whether the census data are missing and a screen has been used. The "OLS" columns use these as regressors. All specifications include an interaction for the sex being missing and a blind audition; "Other covariates" include automatic placement, years since last audition, number of auditions attended, whether a "Big Five" orchestra, size of the audition round, proportion female at the audition round, whether a principal or substitute position, and a dummy indicating whether years since last audition and automatic audition are missing. These are the same specifications as in columns (2), (4), (6), and (8) of Table 6 and column (2) of Table 7. The sample sizes change because in the even-numbered columns we simply

Table 8. Linear Probability Estimates of the Likelihood of Being Advanced Addressing Sex Misclassification

Interpretation. The authors use census-based sex probabilities and instrumental-variable versions to address imperfect sex classification. The broad pattern remains similar, suggesting that name-based sex classification errors are unlikely to drive the findings.

10.7 Table 9. Linear Probability Estimates of the Effect of Blind Auditions on Being Hired with Individual Fixed Effects; Table 10. Probit Estimates of the Effect of Blind Auditions on the Sex of New Members, 1970 to 1996

TABLE 9—LINEAR PROBABILITY ESTIMATES OF THE EFFECT OF BLIND AUDITIONS ON THE LIKELIHOOD OF BEING HIRED WITH INDIVIDUAL FIXED EFFECTS

	Without semifinals		With semifinals		All	
	(1)	(2)	(3)	(4)	(5)	(6)
Completely blind audition	−0.024	0.047	0.001	0.006	0.001	0.005
	(0.028)	(0.041)	(0.009)	(0.011)	(0.008)	(0.009)
Completely blind audition × female	0.051	0.036	0.001	−0.004	0.011	0.006
	(0.046)	(0.048)	(0.016)	(0.016)	(0.013)	(0.013)
Year effects?	No	Yes	No	Yes	No	Yes
Other covariates?	No	Yes	No	Yes	No	Yes
R^2	0.855	0.868	0.692	0.707	0.678	0.691
Number of observations	4,108	4,108	5,883	5,883	9,991	9,991

Notes: The unit of observation is a person-round. The dependent variable is 1 if the individual is advanced (or hired) from the final round and 0 if not. Standard errors are in parentheses. All specifications include individual fixed effects, whether the sex is missing, and an interaction for sex being missing and a completely blind audition. "Other covariates" are the size of the audition, the proportion female at the audition, the number of individuals advanced (hired), whether a "Big Five" orchestra, the number of previous auditions, and whether the individual had an automatic semifinal or final.
Source: Eight-orchestra audition sample. See text.

Table 9. Linear Probability Estimates of the Effect of Blind Auditions on the Likelihood of Being Hired with Individual Fixed Effects

Interpretation. Completely blind auditions raise women's hiring probability mainly in auditions without semifinals. The estimates are economically important but statistically imprecise because hiring is rare and fully blind auditions are limited.

TABLE 10—PROBIT ESTIMATES OF THE EFFECT OF BLIND AUDITIONS ON THE SEX OF NEW MEMBERS: 1970 TO 1996

	Any blind auditions	Only blind preliminaries and/or semifinals vs. completely blind auditions
	(1)	(2)
Any blind auditions	0.238	
	(0.183)	
	[0.075]	
Only blind preliminaries and/or semifinals		0.232
		(0.184)
		[0.074]
Completely blind auditions		0.361
		(0.438)
		[0.127]
Section:		
Woodwinds	−0.187	−0.188
	(0.114)	(0.114)
	[−0.058]	[−0.058]
Brass	−1.239	−1.237
	(0.157)	(0.157)
	[−0.284]	[−0.284]
Percussion	−1.162	−1.164
	(0.305)	(0.305)
	[−0.235]	[−0.235]
p-value of test: only blind preliminaries and/or semifinals = completely blind		0.756
pseudo R^2	0.106	0.106
Number of observations	1,128	1,128

Notes: The dependent variable is 1 if the individual is female and 0 if male. Standard errors are in parentheses. All specifications include orchestra fixed effects and orchestra-specific time trends. Changes in probabilities are in brackets; see text for an explanation of how they are calculated. New members are those who enter the orchestra for the first time. Returning members are not considered new. The omitted section is strings.
Source: Eleven-orchestra roster sample. See text.

Table 10. Probit Estimates of the Effect of Blind Auditions on the Sex of New Members, 1970 to 1996

Interpretation. Roster-based estimates show that blind audition regimes are associated with a higher probability that a new hire is female. The implied probability changes are economically meaningful and help quantify the long-run contribution of blind auditions to orchestra composition.

10.8 Table A1. Sample Descriptive Statistics, Audition Data; Table A2. Sample Descriptive Statistics, Roster Data, 1970 to 1996

Appendix

Table A1—Sample Descriptive Statistics, Audition Data

	Preliminaries							
	Without semifinals		With semifinals		Semifinals		Finals	
	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation	Mean	Standard deviation
.dvanced	0.184	0.387	0.185	0.388	0.349	0.477	0.200	0.400
lind	0.793	0.405	0.976	0.152	0.808	0.394	0.122	0.328
emale	0.376	0.485	0.374	0.484	0.410	0.492	0.411	0.492
emale × Blind	0.305	0.461	0.362	0.481	0.325	0.469	0.056	0.230
lissing female	0.002	0.047	0.002	0.047	0.004	0.066	0	0
lissing female × Blind	0.002	0.043	0.002	0.047	0.004	0.061	0	0
'ears since last audition	2.480	1.661	2.621	2.209	2.432	2.393	2.272	1.895
'ears since last audition, missing	0.663	0.473	0.505	0.500	0.386	0.487	0.505	0.500
.utomatic placement	—	—	—	—	0.267	0.443	0.137	0.345
'umber of auditions attended	1.611	1.137	2.147	1.717	2.490	1.886	2.051	1.513
Big Five" orchestra	0.607	0.488	0.323	0.467	0.213	0.409	0.391	0.488
otal number of auditioners	44.348	22.202	64.279	35.914	15.054	7.187	8.622	4.445
roportion female at round	0.375	0.206	0.373	0.239	0.407	0.211	0.411	0.213
rincipal	0.192	0.394	0.368	0.482	0.353	0.478	0.278	0.448
ubstitute	0.025	0.157	0.005	0.071	0.010	0.101	0.021	0.141
'umber of observations (person-rounds)	5,395		6,239		1,360		1,127	

ource: Eight-orchestra audition sample. See text.

Table A1. Sample Descriptive Statistics, Audition Data

Interpretation. The appendix table summarizes the audition variables by round. It helps show why some estimates are more precise than others: preliminary rounds have many more observations than finals.

Table A2—Sample Descriptive Statistics, Roster Data: 1970 to 1996

	Mean	Standard deviation
portion female among new hires	0.293	0.455
oportion female)$_{t-1}$	0.179	0.081
ly blind preliminary auditions	0.572	0.495
auditions blind	0.104	0.305
:tion:		
Strings	0.642	0.480
Voodwinds	0.158	0.365
Brass	0.165	0.371
Percussion	0.035	0.185
mber of observations	1,128	

'e: Means are musician weighted, not audition weighted.
urce: Eleven-orchestra roster sample. See text.

Table A2. Sample Descriptive Statistics, Roster Data: 1970 to 1996

Interpretation. The roster sample has 1,128 new members, a mean female share among new hires of 0.293, and substantial variation in blind audition regimes across orchestras and years.

10.9 Table A3. Linear Probability Estimates of the Likelihood of Being Advanced by Round; Table A4. Linear Probability Estimates Assessing Short-Panel Bias

	Without semifinals		With semifinals		Semifinals		Finals	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Female	0.007	0.011	−0.054	−0.085	0.103	0.099	0.002	0.0004
	(0.025)	(0.025)	(0.069)	(0.069)	(0.061)	(0.061)	(0.028)	(0.028)
Female × Blind	−0.062	−0.067	0.005	0.037	−0.142	−0.137	−0.091	−0.078
	(0.028)	(0.028)	(0.070)	(0.070)	(0.066)	(0.067)	(0.075)	(0.075)
Blind audition	0.015	0.040	0.024	0.027	0.053	0.115	0.058	0.123
	(0.022)	(0.030)	(0.057)	(0.062)	(0.049)	(0.078)	(0.058)	(0.089)
p-value of H_0: Female + (Female × Blind) = 0	0.000	0.000	0.000	0.000	0.210	0.222	0.207	0.271
Other covariates?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Instrument fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Orchestra fixed effects?	No	Yes	No	Yes	No	Yes	No	Yes
R^2	0.062	0.070	0.033	0.045	0.074	0.081	0.064	0.068
Number of observations (person-rounds)	5,395	5,395	6,239	6,239	1,360	1,360	1,127	1,127

Notes: The dependent variable is 1 if the individual is advanced to the next round and 0 if not. Standard errors are in parentheses. All specifications include dummies indicating whether the sex is missing, and an interaction for the sex being missing and a blind audition. "Other covariates" include automatic round, number of auditions attended, whether a "Big Five" orchestra, size of round, proportion female at the round, and whether a principal (including assistant and associate principal) or substitute position; except in columns (2), (4), (6), and (8) for which "Other covariates" include only automatic placement and number of auditions attended. These results are comparable to those in Table 6 but without individual fixed effects.
Source: Eight-orchestra audition sample. See text.

TABLE A4—LINEAR PROBABILITY ESTIMATES OF THE LIKELIHOOD OF BEING ADVANCED: ASSESSING SHORT-PANEL BIAS

	Part A: Preliminary rounds			
	Preliminaries			
	Without semifinals		With semifinals	
	I[a]	II[b]	I[a]	II[b]
lind	−0.024	−0.042	−0.047	−0.095
	(0.066)	(0.063)	(0.383)	(0.744)
:male × Blind	0.126	0.095	−0.035	0.041
	(0.095)	(0.071)	(0.403)	(0.275)
·value of H_0: Blind + (Female × Blind) = 0	0.233	0.502	0.807	0.943
ther covariates?	Yes	Yes	Yes	Yes
dividual fixed effects?	Yes	Yes	Yes	Yes
ear fixed effects?	Yes	Yes	Yes	Yes
2	0.491	0.537	0.423	0.732
umber of observations (person-rounds)	1,025	639	1,928	55

	Part B: Semifinals and finals, and with orchestra fixed effects					
	Semifinals		Finals		With orchestras fixed effects	
	I[a]	II[b]	I[a]	II[b]	I[a]	II[b]
lind	0.060	0.169	0.123	−0.140	0.084	0.352
	(0.133)	(0.109)	(0.356)	(0.449)	(0.047)	(0.056)
:male × Blind	−0.179	−0.284	0.157	0.403	0.042	0.021
	(0.195)	(0.142)	(0.408)	(0.415)	(0.051)	(0.041)
·value of H_0: Blind + (Female × Blind) = 0	0.438	0.298	0.212	0.587	0.011	0.000
ther covariates?	Yes	Yes	Yes	Yes	Yes	Yes
dividual fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
ear fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
2	0.438	0.593	0.721	0.728	0.506	0.603
umber of observations (person-rounds)	269	223	127	67	2,321	1,776

otes: The dependent variable is 1 if the individual is advanced to the next round and 0 if not. Standard errors are in ırentheses. These are the same specifications as in columns (2), (4), (6), and (8) of Table 6 and column (2) of Table 7.
ɔurce: Eight-orchestra audition sample. See text.
[a] Includes those who auditioned at least three times (for the relevant round).
[b] Includes those who auditioned at least once in a blind audition and at least once in a not-blind audition (for the relevant ›und).

REFERENCES

Becker, Gary. *The economics of discrimination,*

Table A3. Linear Probability Estimates of the Likelihood of Being Advanced by Round

Interpretation. Without individual fixed effects, the female-by-blind interaction can differ sharply from the fixed-effects estimates. This table reinforces why individual-level controls for ability are central to the paper.

Table A4. Linear Probability Estimates of the Likelihood of Being Advanced Assessing Short-Panel Bias

Interpretation. Restricting to musicians observed multiple times or under both blind and not-blind regimes yields results similar in magnitude to the main specifications, reducing the concern that the short panel alone drives the findings.

11 Detailed reading guide

11.1 How to read the coefficients

A coefficient of 0.125 on Female × Blind means that the screen raises a woman's advancement probability by 12.5 percentage points relative to the effect of the screen for men, holding the included controls fixed. It does not mean that all women advance with probability 12.5 percent. The level effect depends on the blind main effect, round, and other covariates.

For example, in Table 6, preliminary rounds without semifinals have a female-by-blind coefficient of 0.125 in column (1) and 0.111 in column (2). These estimates are large relative to the average advancement rates in Table 5.

11.2 Why the paper separates rounds

The audition process is sequential. Preliminary rounds screen out many candidates; semifinal rounds refine the set of acceptable candidates; final rounds select a winner. The screen may matter differently at each step because the committee's objective and information set differ across rounds.

This is exactly what the paper finds. Preliminary and final effects are generally positive for women, while semifinal effects are negative.

11.3 Why the raw and fixed-effects results differ

The raw results reflect both treatment and selection. If blind auditions brought in more female candidates who previously would not have entered, the average female candidate in blind auditions could look less successful even if the screen made evaluation more impartial. Individual fixed effects reduce this problem by comparing the same musician across regimes.

11.4 How the roster and audition samples complement each other

The audition sample observes the mechanism: advancement and hiring decisions inside auditions. The roster sample observes the outcome of interest over a longer horizon: the sex composition of new hires and of orchestra membership. The two samples support the same broad conclusion but answer different parts of the question.

12 Key equations and derivations

12.1 Difference-in-differences interpretation

Ignore controls for a moment and define four conditional advancement probabilities:

$$p_{FB} = E[P \mid F = 1, B = 1], \quad p_{FN} = E[P \mid F = 1, B = 0],$$

$$p_{MB} = E[P \mid F = 0, B = 1], \quad p_{MN} = E[P \mid F = 0, B = 0].$$

The female-specific relative effect of the screen is

$$\Delta = (p_{FB} - p_{FN}) - (p_{MB} - p_{MN}).$$

In the linear model, Δ is the coefficient on $F_i \times B_{jtr}$. This is why the interaction is the key coefficient throughout the paper.

12.2 Why individual fixed effects absorb female status

With individual fixed effects, the equation is

$$P_{ijtr} = \alpha_i + \gamma B_{jtr} + \delta(F_i B_{jtr}) + \cdots + \varepsilon_{ijtr}.$$

Since F_i does not vary within a musician, the separate female coefficient cannot be estimated. But $F_i B_{jtr}$ varies for women when they appear in both blind and not-blind rounds. Therefore δ is still identified.

12.3 Decomposition of female new-hire growth

Let p be the fraction of candidates who are women, s_F the success probability for female candidates, and s_M the success probability for male candidates. If an audition has n candidates, the expected number of female hires is proportional to

$$nps_F.$$

The female share among hires is

$$\frac{ps_F}{ps_F + (1-p)s_M}.$$

The screen can raise this share through s_F, while broader social and educational changes can raise it through p. The paper argues that both channels matter.

13 Bottom line

Goldin and Rouse provide evidence that blind auditions made elite orchestra hiring more impartial. The paper's strongest contribution is not just showing that more women were hired after 1970, but connecting the change to a specific procedural intervention and using individual-level audition records to control for musician fixed effects.

The final conclusion is nuanced. Not every coefficient is precise, and the semifinal results go in the opposite direction. But the combined evidence from raw matched comparisons, individual fixed-effects regressions, robustness checks, and roster data indicates that screens increased women's success and helped transform the composition of U.S. symphony orchestras.